

tured, usually wears a body suit with reflective markers taped or glued to specific points all over his or her body. Several cameras film the subject as he or she performs specific motions—for example, the multiple Mr Smiths in *The Matrix Revolutions* were created by mapping actor Hugo Weaving's face and then transposing it on to the CG replicas.

The cameras report the exact position of each reflective marker, hundreds of times a second, to a computer. A magnetic system can also be used in place of the optical system—a sensor detects a nearby magnetic field and transmits data on its 3D position to the computer. Camera motions are also fed into the computer. MoCap software then processes this data to determine the exact movement of the subject. The animation data exported by the computer, can be manipulated using software such as Maya or 3D Studio Max.

MoCap in India

The word 'animation', a decade ago, would invoke thoughts of cartoon characters—remember Bugs Bunny? Today, realism is key, and movies use animation techniques such as MoCap to achieve this realism in tales of fantasy. MoCap is also used in computer games and television shows.

In India, very few animation studios have invested in MoCap systems, mainly due to the absurdly high costs of the equipment. Pentamedia (Chennai) and FX Factory (Mumbai) are two studios that have spent the big bucks on this equipment.

Chennai-based Pentamedia have one of the largest Motion Capture facilities in the world. The company has produced full-length animation films (including *Simbad—Beyond The Veil Of Mists*; *Pandavas—The Five Warriors*; *Alibaba and Son of Aladdin*), and also specialises in SFX for Tamil (Shankar's *Jeans*) and other regional language films.

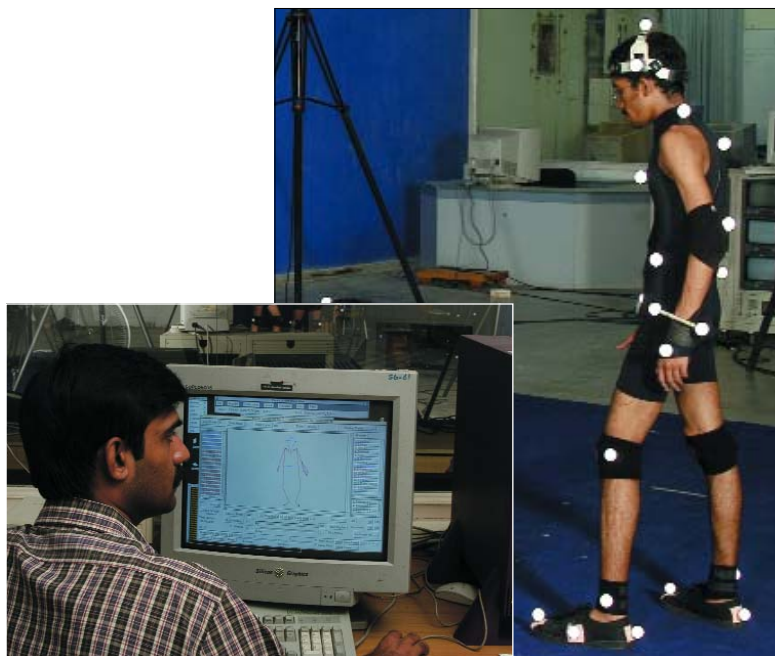
Pentamedia's setup includes high-speed Falcon cameras, capable of running between speeds of 60 to 240 frames per second (fps). The cameras read data from special scotch-brite type markers that are fitted onto a subject's body. The cameras can capture data from anywhere within the 20 x 20 ft capture area, and software such as 3D Studio Max, Maya and Soft Image can use this data to create the required animation.

Riyaz Babu, who manages the 80'x 40'x 80' Mo Cap facility at Pentamedia says, "Our Optical Mo Cap system enables data capture from multiple characters, animals, stunt and sports sequences as well."

FX Factory's 50-member team of animators, art directors, storyboard artists, sculptors and visual effects supervisors is led by special-effects expert Ramesh Meer. They use full-body motion capture systems—Gypsy 2.5 and Gypsy 3.0, manufactured by Saicom Studio Systems. Gypsy is a cost-effective MoCap system, and has a large wireless capture range—up to half a mile (800 m) outdoors and up to 200 yards (180 m) indoors. This system does not require cameras, and records motion directly from the body. It is also portable, and works with any PC.

Fantasyland

The character 'Gollum', in Peter Jackson's trilogy *The Lord of the Rings* (LOTR), was filmed using MoCap. Oscar-winning director of photography



MoCap equipment helps animators study movement precisely

Origins

MoCap probably takes its origin from rotoscoping, a process that involves creating a series of drawings by projecting individual frames of an actor's filmed performance onto sheets of paper. These drawings are then filmed to create an animated sequence, which can be as lifelike as possible, in appearance and movement. Disney Studios used this method to create the human characters in *Snow White*—the animation was traced out over film footage of live actors playing out the scenes.

Andrew Lesnie and his team first captured actor Andy Serkis' performance on the set. Animators then used his movements and facial expressions, captured with MoCap cameras, and applied them to the animated Gollum. Motion capture photography, CG imagery and digital sound mixing combined to produce what you saw in *LOTR*.

MoCap starred in George Lucas' *Star Wars - Episode II Attack of the Clones*, especially in the form of Jar Jar Binks—a completely CG character, played in costume by Ahmed Best. Jar Jar, incidentally, was the first CG character in Hollywood filmdom ever to interact realistically with live actors.

MoCap has been used to create mind-blowing SFX in films like *Titanic* (1997), *Gladiator* (2000) and, of course, *The Matrix* series. Even full-length animation films such as Columbia Pictures' *Final Fantasy: The Spirit Within* are believed to have used MoCap data.

Hotshot director John Woo is currently exploring the possibility of translating Nintendo's best-selling video game franchise 'Metroid' onto the big screen—large portions of which will involve MoCap. Robert Zemeckis' *The Polar Express* (due for release in 2005) plans to use MoCap to create an animated character 'played' by Tom Hanks.

MoCap is also used to create games such as Electronic Arts' *FIFA* and *Cricket* series, games based on the *LOTR* trilogy, and *James Bond 007: Everything or Nothing*, in which actor Pierce Brosnan's expressions were digitally mapped. *James Bond: 007 Nightfire*, a cinematic simulation of the original Bond game used MoCap for half the project—the rest was animated by hand.

Bangalore-based Dhruva Interactive, a game developer, was the first Indian company to be signed on by a major global publisher to develop an international title—*Mission: Impossible PC*, in 1999. Says CEO Rajesh Rao, "We used MoCap stu-